

Core Architecture Concepts

- **Browser Hierarchy (Browser $\rightarrow$ Context $\rightarrow$ Page):**
Playwright's isolation model. A *Browser* is the core engine; a *Context* is an isolated incognito session that guarantees zero data contamination between runs; a *Page* is an active tab where rendering and interaction take place.
- **HTML DOM (Document Object Model):** The structural tree-like blueprint of a webpage. Browser automation frameworks read this data layer rather than relying on visual pixel coordinates.
- **CSS Selectors / Locators (#id):** Unique geometric text anchors within the DOM code that point directly to specific user interface elements (like text fields or buttons) so automation engines can target them with zero ambiguity.
- **Headed vs. Headless Execution:** Headed mode opens a visible graphical browser window on the desktop for debugging. Headless mode executes the browser purely within background system memory, maximizing speed and minimizing resource draw.